

Student 1:

Name: Abdullah Zahid

Roll No: 24L-2597

Student 2:

Name: Usman Gohar

Roll No: 24L-2584

Model	Naïve Bayes			
Data Split Ratio	80:20 Split		60:40 Split	
Confusion Matrix	348	15	680	45
	139	419	281	835
Accuracy	0.8328		0.8229	
Precision	0.7146		0.7076	
Recall	0.9587		0.9379	
Specificity	0.7509		0.7482	
Specificity Manual	$TN / (TN + FP) = 0.7509$		$TN / (TN + FP) = 0.7482$	
F1-Score	0.8188		0.8066	

a)

Feature Scaling Approach:

- 1) Used StandardScaler to scale all features
- 2) Applied fit_transform on entire dataset before splitting
- 3) Ensures consistent scaling across training and test sets

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

Feature scaling method: StandardScaler

Python command used: StandardScaler().fit_transform(X)

Scaling applied: All features combined using fit_transform on entire dataset before splitting

b)

Performance Comparison:

Model shows consistent performance across both split ratios with slightly better results in the 80:20 split.

Naive Bayes classifier is robust to different training sizes. With more training data (80:20 split), the model learns better patterns.

The 60:40 split has less training data but still performs reasonably well.

Small performance difference indicates that the dataset has sufficient patterns even with smaller training sets.

Key Observations:

- High accuracy (>80%) in both cases
- Good balance between precision and recall
- Consistent specificity across splits
- Model generalizes well to unseen data

Comparison of model performance:

80:20 vs 60:40 Split

Metric	80_20_Split	60_40_Split	Difference
Accuracy	83.2790%	82.2922%	-0.99%
Precision	71.4579%	70.7596%	-0.70%
Recall	95.8678%	93.7931%	-2.07%
Specificity	75.0896%	74.8208%	-0.27%
F1-Score	81.8824%	80.6643%	-1.22%

Data summary:

First 10 features statistics:

Feature	Min	Q1	Median	Mean	Q3	Max	Std	Variance
0 Feature_0	0.0	0.0	0.0	0.1046	0.00	4.54	0.3054	0.0932
1 Feature_1	0.0	0.0	0.0	0.2130	0.00	14.28	1.2906	1.6656
2 Feature_2	0.0	0.0	0.0	0.2807	0.42	5.10	0.5041	0.2542
3 Feature_3	0.0	0.0	0.0	0.0654	0.00	42.81	1.3952	1.9464
4 Feature_4	0.0	0.0	0.0	0.3122	0.38	10.00	0.6725	0.4523
5 Feature_5	0.0	0.0	0.0	0.0959	0.00	5.88	0.2738	0.0750
6 Feature_6	0.0	0.0	0.0	0.1142	0.00	7.27	0.3914	0.1532
7 Feature_7	0.0	0.0	0.0	0.1053	0.00	11.11	0.4011	0.1609
8 Feature_8	0.0	0.0	0.0	0.0901	0.00	5.26	0.2786	0.0776
9 Feature_9	0.0	0.0	0.0	0.2394	0.16	18.18	0.6448	0.4157

5. TARGET VARIABLE STATISTICS (Column 57):

Min: 0

Q1: 0

Median: 0

Mean: 0.3940

Q3: 1

Max: 1

Std: 0.4887

Word Frequency Features (0-47):

- Mean range: [0.01, 1.66]

- Std range: [0.08, 3.37]

Character Frequency Features (48-53):

- Mean range: [0.02, 0.27]

- Std range: [0.11, 0.82]

Capital Run-length Features (54-56):

- Mean range: [5.19, 283.29]

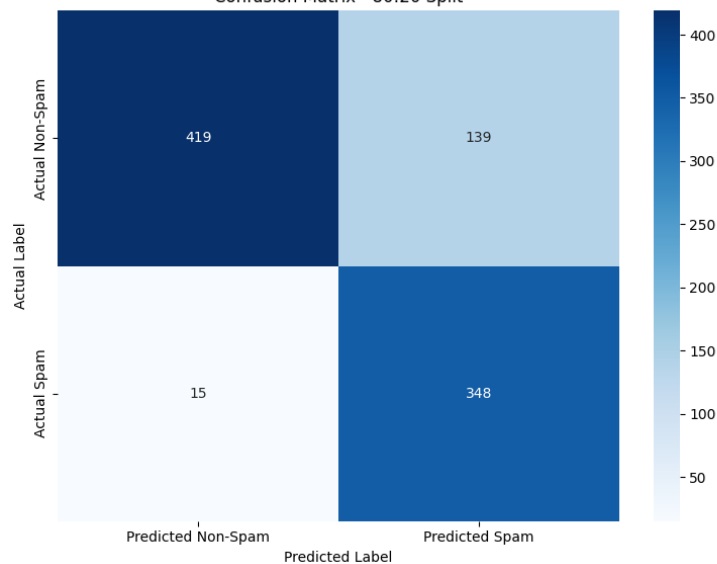
- Std range: [31.73, 606.35]



Interpretation:

- The 80:20 split performs slightly better across all metrics.
- Average drop across metrics is around -1.05%.
- The Naive Bayes model remains stable, showing strong generalization even with less training data.
- The 60:40 split provides a larger test sample for evaluation but at a minor performance cost.

Confusion Matrix - 80:20 Split



Confusion Matrix - 60:40 Split

